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A Systematic Literature Review on How Cybercriminals are Utilizing Deepfakes to Conduct Cyberattacks

Deepfake technology is becoming an increasingly powerful tool for cybercriminals, enabling more convincing and sophisticated cyberattacks. From phishing scams and social engineering to identity fraud and disinformation campaigns, deepfakes manipulate trust in digital communication, leading to financial, psychological, and reputational harm. This systematic literature review examines how deepfakes are being used in cybercrime, the emerging trends, the ethical, legal, and societal challenges they pose, and the gaps in existing research and detection methods.

This review explores three key themes guided by core research questions: the emerging trends in deepfake-enabled cybercrime, the broader implications of these threats, and the current limitations in detection and mitigation. It identifies a growing use of multi-modal deepfakes that combine synthetic video, audio, and text, as well as real-time deepfake manipulation that bypasses traditional security measures. It also highlights concerns around privacy violations, reputational damage, and the increasing difficulty of verifying digital evidence in legal proceedings. A lack of standardized regulations and inconsistent policies across jurisdictions further complicate efforts to counteract these attacks.

The study finds that research on deepfake cybercrime remains limited, with gaps in real-world case studies, weaknesses in current detection technologies, and a need for more advanced forensic tools. Based on these findings, this review offers recommendations for strengthening cybersecurity defenses, improving public awareness, and fostering collaboration between industry, academia, and policymakers.

CCS CONCEPTS• Deepfake • Cybersecurity • Cyber attacks

1 INTRODUCTION

Deepfake technology has rapidly emerged as a disruptive application of artificial intelligence (AI), leveraging deep learning algorithms to create hyper-realistic synthetic media that convincingly mimic individuals' appearances, voices, and behaviors (Mustak et al., 2023; Nightingale & Farid, 2022). While deepfakes offer promising applications in fields such as entertainment, education, and accessibility, their misuse has introduced significant cybersecurity challenges, enabling attackers to impersonate individuals with unprecedented realism and erode trust in digital systems.

The term "*deepfake*" originated in 2017 from a Reddit user who used open-source face-swapping technology to create synthetic videos (Mustak et al., 2023). The term combines "*deep learning*" and "*fake*," reflecting the technology's ability to generate fraudulent content using generative adversarial networks (GANs). These GAN

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ACM 2576-5337/2026/3-ART
https://doi.org/10.1145/3801545

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models consist of two competing networks: a generator that produces synthetic media and a discriminator that evaluates its authenticity. Through iterative refinement, the generator learns to create content indistinguishable from real images, videos, and audio (Blancaflor et al., 2024; Nightingale & Farid, 2022).

Recent advancements have significantly expanded the capabilities of deepfakes, with innovations such as real-time manipulation, multi-modal synthesis, and low-resolution attacks complicating detection efforts. Tools like *DeepFaceLab* and GAN-based platforms now enable real-time impersonation, enhancing the effectiveness of social engineering attacks, including business email compromise (BEC), voice phishing (vishing), and synthetic identity fraud (Gambín et al., 2024; N. A. Karim et al., 2024). The rise of multi-modal deepfakes, which combine synthetic video, audio, and text, further amplifies the threat landscape by making phishing schemes and disinformation campaigns more convincing and harder to identify (Malinka, Firc, Kaška, et al., 2024). Deepfakes have also become embedded in wider cyber risk ecosystems. They introduce new layers of deception, exposing individuals and organizations to heightened risks through novel wetware hacks. As noted in a recent study, these attacks increasingly target not only individuals but also critical infrastructure, financial institutions, and democratic processes (Maras & Logie, 2024). The accessibility of user-friendly deepfake tools has further empowered threat actors, reducing the technical expertise required to conduct sophisticated attacks (Cross, 2022).

Beyond technical concerns, the misuse of deepfake technology raises complex ethical, legal, and societal issues. These include privacy violations, non-consensual pornography, gender-based violence, and reputational sabotage, with marginalized groups often facing the most severe consequences (Mustak et al., 2023). The limitations of cohesive international regulatory frameworks worsen these challenges, as existing legislation often fails to address the cross-border nature of deepfake-driven cybercrime (Songja et al., 2024; Zheng et al., 2025).

Although deepfake generation and detection have attracted significant research interest, current scholarship on deepfake-enabled cyberattacks remains fragmented across technical, legal, psychological, and organizational domains (Juefei-Xu et al., 2022; Songja et al., 2024). Existing reviews focus predominantly on media forensics and detection methods, offering limited synthesis of how deepfakes are operationalized within cybercrime or how they interact with broader socio-technical systems. This fragmentation is becoming more consequential as generative AI (GenAI) tools, particularly diffusion models and multimodal architectures, rapidly expand access to highly sophisticated synthetic media. Understanding how cybercriminals operationalize deepfake technology therefore requires a comprehensive and integrative synthesis rather than further isolated case studies or purely technical evaluations.

While earlier reviews have examined detection, authenticity verification, or specific attack types, no existing work provides an integrated assessment that connects attacker behaviors, socio-technical vulnerabilities, regulatory challenges, and emerging multimodal threat patterns. This review fills that gap by offering the first holistic synthesis that links technical advances with operational misuse and broader societal risks, thereby clarifying what is known, where critical blind spots remain, and how the field should adapt to the evolving deepfake threat landscape.

To accomplish this, the review categorizes deepfake-driven cyberattacks, evaluates current detection capabilities, and examines adaptive defense mechanisms, including blockchain-integrated authentication frameworks, attention-based convolutional long short-term memory (LSTM) networks, and multi-collaborative GANs (MCGAN)(Gambín et al., 2024; N. A. Karim et al., 2024). It also highlights how deepfakes intersect with other cyber threats, expanding the scale and impact of attacks while raising broader societal and regulatory concerns.

We identify key research gaps in deepfake-enabled cybercrime, including the emerging trends in how cybercriminals leverage deepfakes, the ethical, legal, and societal challenges they create, and the limitations in current detection and mitigation strategies. By situating these findings within broader developments in GenAI, this review clarifies where the field has progressed, where foundational gaps remain, and where future research should focus to support more resilient cybersecurity, policy, and forensic responses.

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2 METHODOLOGY

To ensure a comprehensive and methodologically rigorous review, this study adhered to the *PRISMA 2020* guidelines (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)(Page et al., 2021). This framework was selected for its established reliability in enhancing the transparency and reproducibility of systematic reviews, particularly when synthesizing evidence from diverse sources. Given the rapidly evolving nature of deepfake technology and the multifaceted landscape of cybersecurity, the PRISMA guidelines were instrumental in standardizing the review process (Kazimierczak et al., 2024). They provided a structured approach for systematically identifying, screening, and including relevant studies, thereby reducing the risk of selection bias and ensuring that the review captures key insights across technical, ethical, and legal domains. By following this rigorous framework, our review achieves a higher degree of replicability, clarity in study selection, and consistency in reporting emerging trends and research gaps.

To guide the review to understand how deepfakes are being used in cyberattacks, the following research questions (RQs) were developed:

RQ 1. What are the emerging trends in cybercrime involving the use of deepfakes?

RQ 2. What are the ethical, legal, and societal implications of deepfake-driven cyberattacks?

RQ 3. What gaps exist in current research on deepfake-enabled cyberattacks, and what challenges remain in understanding and addressing them?

The literature search was conducted across five major academic databases: *ACM Digital Library*, *IEEE Xplore*, *ScienceDirect*, *Scopus*, and *SpringerLink*. These sources were selected for their strong coverage of research on deepfake technology and cybersecurity. The review considered peer-reviewed journal articles and conference papers published from January 2017, when the term deepfake was introduced, to January 2025. The following four-stage search process was then defined to ensure the inclusion of studies on deepfake applications in cybercrime:

- **Identification:** Relevant studies were retrieved from the databases, and duplicate records were removed using Zotero, an open-source reference management software designed to help researchers collect, organize, cite, and share research materials (S. Kaur & Dhindsa, 2016).
- **Screening:** Titles, abstracts, and keywords were reviewed to exclude studies focused on entertainment or unrelated AI applications.
- **Eligibility:** Full text assessments determined whether studies met the inclusion criteria. Only peer reviewed research published in English between 2017 and 2025 that directly addressed deepfake related cybercrime was considered. Opinion pieces and studies lacking methodological rigor were excluded.
- **Inclusion:** The final selection consisted of high-quality studies that provided meaningful insights into the cybersecurity implications of deepfake technology.

Key information extracted from each study included title, authors, year of publication, country, research objectives, methodology, findings, and limitations. Thematic analysis was used to categorize findings into three main themes: *emerging trends*, *ethical, legal, and societal implications*, and *research gaps*. Figure 1 provides an overview of the key themes identified in the literature, while Figure 2 outlines the SLR process.

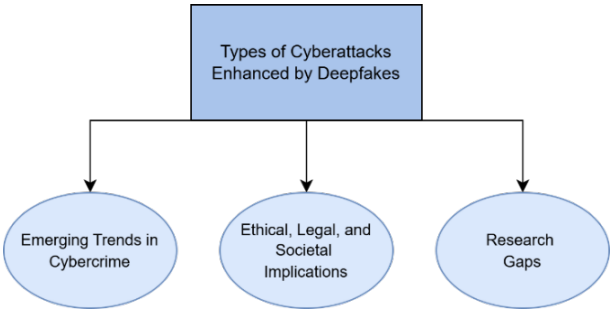


Figure 1: Figure 1 Key Themes in the Literature

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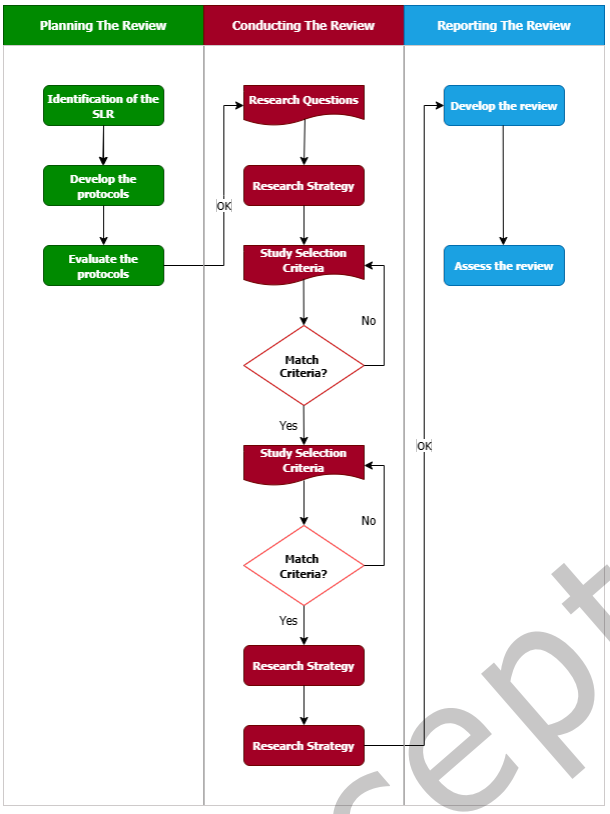


Figure 2: The process of the SLR

Table 1 outlines the stages of the SLR process, detailing the methods used to identify, screen, and select relevant studies.

Table 1: The stages of the SLR

Stage	Description	Purpose
Identification	Conducted a thorough search across five databases: ACM Digital Library, IEEE Xplore, ScienceDirect, Scopus, and SpringerLink.	Capture relevant, high-impact research in deepfake and cybersecurity domains.
Screening	Reviewed titles, abstracts, and keywords while removing duplicates using Zotero.	Narrow down to studies directly pertinent to deepfake and cybersecurity.
Eligibility	Conducted full-text assessments based on inclusion criteria, focusing on papers published in English from 2017 onward.	Ensure the relevance, quality, and rigor of included studies.
Inclusion	Final selection of high-quality, peer-reviewed studies meeting all criteria.	Provide a comprehensive, high-quality overview of current research.

2.1 Search Strategy and Keywords

The initial process yielded 2302 results, distributed as follows: ACM Digital Library (251), IEEE Xplore (274), ScienceDirect (786), Scopus (296), and SpringerLink (695). After removing duplicates, 1725 unique papers remained. Boolean operators were used to balance inclusivity and specificity across databases. The search string below was used for ACM Digital Library, IEEE Xplore, Scopus, and SpringerLink.

Search String: *("Deepfake" OR "Deep fake" OR "Deepfakes" OR "Deep fakes") AND ("Cybersecurity" OR "Cyber security" OR "Cyber-attacks" OR "Cyber attacks" OR "Cybercrime" OR "Cyber crime" OR "Cyber threats" OR "Cyber threat" OR "Cyber intrusions" OR "Cyber intrusion" OR "Phishing" OR "Phishing attacks" OR "Phishing attack" OR "Social engineering" OR "Social engineering attacks" OR "Social engineering attack" OR "Identity theft" OR "Identity thefts" OR "Cybersecurity breaches" OR "Cyber security breaches" OR "Cybersecurity breach" OR "Cyber security breach" OR "Threat detection" OR "Threat detections" OR "Fraud" OR "Online fraud" OR "Digital fraud")*

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Due to ScienceDirect's restriction of eight Boolean operators per search, the search was broken down into five smaller queries:

First Query: *("Deepfake" OR "Deep fake" OR "Deepfakes" OR "Deep fakes") AND ("Cybersecurity" OR "Cyber security" OR "Cyber-attacks" OR "Cyber attacks")*

Second Query: *("Deepfake" OR "Deep fake" OR "Deepfakes" OR "Deep fakes") AND ("Cybercrime" OR "Cyber crime" OR "Cyber threats" OR "Cyber threat")*

Third Query: *("Deepfake" OR "Deep fake" OR "Deepfakes" OR "Deep fakes") AND ("Cyber intrusions" OR "Cyber intrusion" OR "Phishing" OR "Phishing attacks")*

Fourth Query: *("Deepfake" OR "Deep fake" OR "Deepfakes" OR "Deep fakes") AND ("Social engineering" OR "Social engineering attacks" OR "Identity theft" OR "Identity thefts")*

Fifth Query: *("Deepfake" OR "Deep fake" OR "Deepfakes" OR "Deep fakes") AND ("Cybersecurity breaches" OR "Cyber security breaches" OR "Fraud" OR "Online fraud")*

A second screening phase involved a detailed full text review to confirm that selected studies directly addressed the core research questions.

The secondary screening applied the following inclusion criteria:

- Papers explicitly discussing deepfakes in cyberattacks or cybercrime, including fraud, identity theft, social engineering, and cybersecurity breaches.
- Detection focused studies only if they directly linked deepfakes to malicious activities such as phishing, fraud, or identity theft.
- Case studies or real-world examples of deepfakes used for cybercrime, ensuring practical relevance.

This targeted approach resulted in **188** studies which were directly relevant to how cybercriminals exploit deepfakes. Each paper underwent a critical evaluation of methodology, findings, and discussion to confirm relevance and quality. Table 2 shows the results of the screening and selection process.

Table 2: Results of Screening and Selection

Stage	Number of Studies	Description
Database Search	2302	Total studies retrieved from five databases: ACM Digital Library (251), IEEE Xplore (274), ScienceDirect (786), Scopus (296), and SpringerLink (696). The search focused on titles, keywords, and abstracts.
Duplicates Removed	1725	Unique studies remaining after removing duplicate entries using Zotero reference management software.
Further Screening	188	Studies classified as directly relevant after an in-depth review of abstracts, introductions, and conclusions. This phase focused on papers discussing deepfake-enhanced cyberattacks, including real-world case studies and malicious applications.

During the detailed review, a thematic analysis was conducted to identify case studies, trends, and patterns at the intersection of deepfakes and cyberattacks. Coding was guided by the three research questions and focused on recurring concepts related to attack modalities, detection limitations, legal and ethical issues, victim experiences, and organizational responses. These codes were iteratively grouped into higher-level categories, allowing us to compare how different studies described similar phenomena, where they diverged, and which areas remained underexplored. This process yielded three primary themes: *emerging trends in cybercrime*, *ethical, legal, and societal implications*, and *research gaps*. These themes provide a structured framework for moving beyond description toward an integrated understanding of both the technical and socio-technical dimensions of deepfake-related cybercrime.

2.2 Contributions and Relevance

Our systematic review makes an important contribution to cybersecurity by examining how cybercriminals exploit deepfake technology in conducting cyberattacks. Using a structured approach aligned with PRISMA 2020 guidelines, we analyzed studies from five major academic databases, ensuring a broad interdisciplinary perspective that integrates insights from cybersecurity, law, social sciences, and psychology.

To address how deepfakes are being used in cyberattacks, we identified several emerging trends (RQ1) that demonstrate their increasing role in cybercrime. Our findings highlight the increasing use of deepfakes for real-

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time manipulation, multimodal attacks, and evasion of detection systems (Gambín et al., 2024; Malinka, Firc, Kaška, et al., 2024). Cybercriminals are leveraging synthetic media to bypass biometric authentication, execute business email compromise scams, and carry out impersonation-driven ransomware attacks, ultimately eroding trust in digital communication.

In examining the ethical, legal, and societal implications of deepfake-driven cyberattacks (RQ2), our findings highlight pressing challenges in governance and regulation. While some legal frameworks attempt to criminalize malicious deepfakes, enforcement remains inconsistent, and international collaboration is limited (Zheng et al., 2025). Ethical concerns revolve around consent, data privacy, and the potential misuse of AI-generated content, while the societal impacts include the erosion of public trust, increased susceptibility to misinformation, and complications in digital forensics.

To explore the research gaps in deepfake-enabled cyberattacks (RQ3), we identified several critical shortcomings in the existing literature. Empirical studies on real-world deepfake cyberattacks remain scarce, and current detection methods struggle to keep pace with rapidly advancing GANs (Karim et al., 2024), reinforcing the urgent need for adaptive security frameworks and international collaboration.

By systematically assessing these threats and research gaps, we provide valuable insights that can inform cybersecurity strategies, improve detection methodologies, and shape policy development. Our review also highlights the importance of industry-academic collaboration in strengthening digital security and developing effective countermeasures against deepfake-driven cybercrime.

3 RESULTS

Our analysis of 188 studies provides valuable insights into how cybercriminals exploit deepfake technology. To better reflect these findings, we categorized them into three key themes based on the RQs: *emerging trends in cybercrime*, *ethical, legal, and societal implications*, and *research gaps*.

Table 3 presents key insights from the literature review, categorizing findings into emerging trends in cybercrime, ethical, legal, and societal implications, and research gaps. The quantified findings column provides a data-driven overview, showing the number and percentage of studies addressing each theme. Each key insight is supported by two example studies, which serve as representative references rather than an exhaustive list. This structure highlights both the current state of research and areas requiring further investigation.

Table 3: Insights from Literature Review

Theme	Key Insights	Example Studies	Quantified Findings (Count & Percentage of 188 Studies)	Research Gaps Identified
Emerging Trends in Cybercrime	Deepfakes increasingly enhance phishing, BEC scams, vishing, and social engineering.	(N. A. Karim et al., 2024; Schmitt & Flechais, 2024)	68 studies (37%) reported deepfakes being used in phishing attacks.	Lack of real-world case studies for multi-modal deepfake cyberattacks.
	Real-time manipulation during video calls and synthetic voice for scams.	(Gambín et al., 2024; Malinka et al., 2024)	40 studies (22%) mentioned real-time deepfake manipulation during cyberattacks.	Limited empirical evidence on cross-platform deepfake attacks.
	Increased use of GANs and diffusion models for generating hyper-realistic content.	(Juefei-Xu et al., 2022; Sun et al., 2025)	57 studies (31%) highlighted the role of GANs in advancing deepfake realism.	Need for datasets representing evolving deepfake generation techniques.
Ethical, Legal, and Societal Implications	Deepfake-enabled disinformation campaigns and privacy violations.	(Maras & Logie, 2024; Songja et al., 2024)	51 studies (28%) reported deepfakes used in disinformation and propaganda.	Lack of cohesive international legal frameworks.
	Rise of non-consensual synthetic media and gender-based violence.	(Cross, 2022; Viola & Voto, 2023)	35 studies (19%) addressed non-consensual deepfake	Need for gender-sensitive regulations

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			abuse, predominantly targeting women.	and reporting mechanisms.
	Potential for election interference and manipulation of public opinion.	(R. Kumar et al., 2024; Pawelec, 2022)	26 studies (14%) highlighted concerns regarding election-related deepfakes.	Gaps in understanding regional variations in the impact of deepfakes.
Identified Research Gaps	Limited empirical studies on deepfake-driven cyberattacks across different regions.	(Khan et al., 2023; Sandoval et al., 2024)	17 studies (9%) provided empirical data from non-Western countries.	Need for longitudinal studies to track deepfake threat evolution.
	Insufficient datasets for training and benchmarking detection systems.	(Dagar & Vishwakarma, 2025; Pandey et al., 2024)	22 studies (12%) reported availability of multi-modal datasets for detection.	Lack of cross-cultural datasets and multi-lingual deepfake examples.
	Inadequate understanding of psychological impact on victims.	(Roberts, 2023; Thumboo & Mukherjee, 2024)	15 studies (8%) examined the psychological impact on victims of deepfake attacks.	Scarcity of victim-focused research, particularly in underrepresented areas.

As Figure 3 shows, research interest in deepfake-related cybersecurity threats has grown significantly in recent years. The number of relevant publications increased from 8 in 2020 to 77 in 2024, with 13 papers already identified in the first month of 2025. This surge highlights the rising recognition of deepfakes as a cybersecurity threat and underscores the need for further research, innovative countermeasures, and more robust detection frameworks.

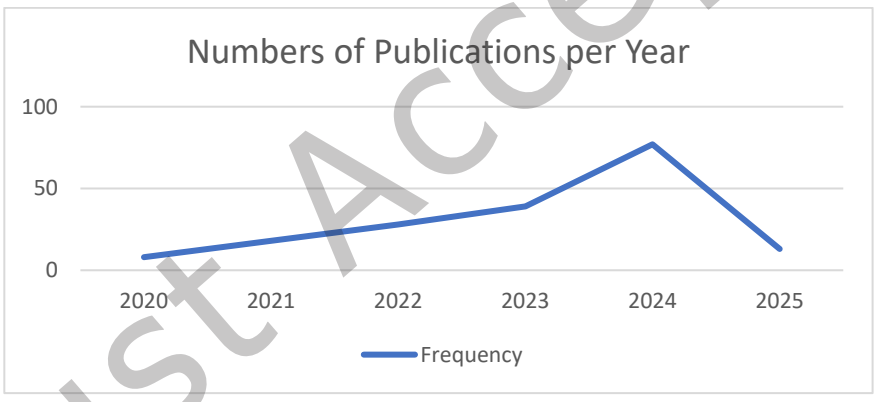


Figure 3: Number of Publications Per Year

Figure 4 shows the distribution of the research methodologies examined in this systematic review reflect the progression of deepfake-related cybersecurity research. Experimental studies form the largest category, emphasizing detection mechanisms, deepfake generation, and security vulnerabilities. Theoretical research explores frameworks for understanding deepfake-enabled cybercrime, though much of the discourse remains centered on defining risks rather than proposing countermeasures. Literature reviews synthesize existing knowledge and highlight detection gaps, while empirical research, including case studies and surveys, remains limited. Mixed-methods studies help bridge theoretical discussions with real-world applications, underscoring the need for interdisciplinary collaboration and more empirical research to develop effective countermeasures.

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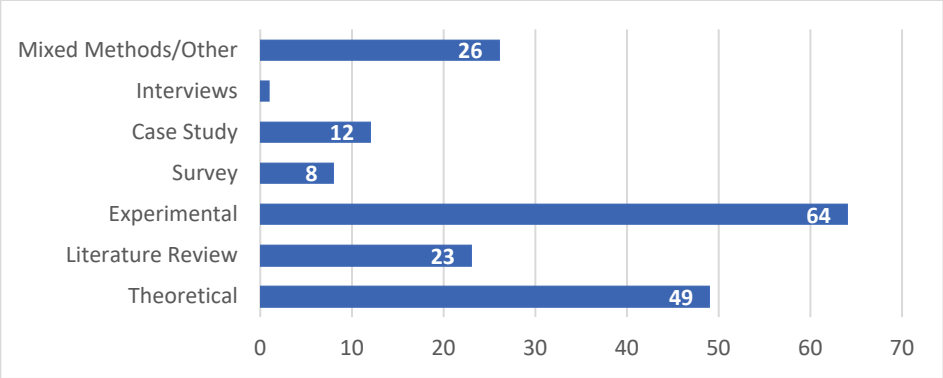


Figure 4: Research Methodologies Identified

3.1 Deepfakes as a Tool for Cyberattacks

Before addressing the RQs in a thematic analysis, we first illustrate the different types of cyberattacks enhanced by deepfakes, as understanding these attack vectors provides context for the broader implications of this technology. The analysis of 188 papers highlights several key attack vectors where deepfakes play a critical role.

We will now illustrate the different types of attacks enhanced by deepfakes, as emerged from our SLR.

Phishing and Social Engineering: Our review illustrated how Deepfake technology significantly enhances phishing and social engineering attacks by creating realistic audio and video content that manipulates victims into divulging sensitive information or performing unauthorized actions. Social engineering attacks exploit psychological manipulation, often leveraging deepfake-generated media to increase credibility and urgency.

Voice Phishing (Vishing): Deepfake audio technology is used to mimic trusted individuals, such as company executives or family members, to manipulate victims into transferring funds or sharing confidential information. Schmitt and Flechais (2024) highlighted how generative AI enhances the effectiveness of social engineering and business email compromise (BEC) tactics through voice cloning. Attackers deploy deepfake audio and video to impersonate executives or high-ranking officials during virtual meetings or email exchanges. These attacks exploit trust, leading to unauthorized financial transactions or the exposure of sensitive data. For example, one documented incident involved a UK-based energy company whose CEO authorized a transfer of approximately $243,000 after receiving a deepfake audio call that convincingly mimicked the voice of a senior executive(Damiani, n.d.)

Video Phishing: Attackers create realistic video messages impersonating trusted figures to deceive recipients into clicking malicious links or sharing credentials. A study emphasized how face synthesis and identity swaps are used for social engineering (Juefei-Xu et al., 2022). These attacks often bypass traditional spam filters and cybersecurity defenses due to the convincing nature of the deepfake content.

Real-Time Manipulation: Advances in GANs now allow for real-time deepfake manipulation during video calls, making social engineering attacks more convincing and difficult to detect. (S. Karim et al., 2024) demonstrated how real-time deepfake manipulation could bypass identity verification during video conferencing, raising concerns for remote work environments.

Identity Theft and Fraud: Deepfake technology facilitates identity theft and fraudulent activities by creating synthetic identities and forging identification documents. The ability to manipulate biometric identifiers poses significant threats to digital identity systems widely used in banking, healthcare, and governmental services (N. A. Karim et al., 2024).

Synthetic Identity Creation: Deepfakes enable the creation of entirely new personas used to bypass identity verification systems, as highlighted in a study on the analysis of romance fraud cases (Cross, 2022). Victims often believe they are interacting with real individuals, making detection challenging for both users and platforms. (Nightingale & Farid, 2022) found that AI-synthesized faces were nearly indistinguishable from real faces, with participants unable to differentiate between authentic and synthetic identities.

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Document Forgery: Deepfake-generated images and videos are used to create fake identification documents for fraudulent transactions and identity theft. (Ali et al., 2022) reported cases where attackers used deepfake videos to impersonate individuals during identity verification processes, often bypassing advanced detection systems. Additionally, Deepfakes are used to alter video or audio evidence, complicating legal investigations and judicial processes. The ability to manipulate digital evidence undermines trust in legal systems and complicates the administration of justice. (Sandoval et al., 2024) highlighted how deepfake-generated evidence can undermine trust in legal proceedings, making it difficult to distinguish between authentic and manipulated content. For instance, deepfake audio recordings have been used to falsify confessions and witness testimonies, complicating criminal investigations and court cases.

Disinformation and Misinformation Campaigns: Deepfakes play a central role in disinformation campaigns, particularly during political elections and geopolitical conflicts. These campaigns often involve the dissemination of manipulated videos and audio recordings designed to spread false narratives, undermine public trust, and influence voter behavior.

Election Manipulation: Deepfake videos and audio recordings are used to spread false information about political candidates, influencing public opinion and undermining democratic processes (Lipińska, 2024; Pawelec, 2022; Pranay Kumar et al., 2024). For example, (Matatov et al., 2022) highlighted cases where deepfake videos of political figures were used to disseminate fabricated statements during election campaigns, exacerbating societal polarization. Analyses of the 2019 political crisis in Gabon have noted that a disputed video of then-President Ali Bongo, widely suspected of being manipulated, contributed to uncertainty about his health and helped catalyze political instability (Cahlan, 2020).

Fake News: Deepfake-generated content is disseminated on social media platforms to spread false narratives, often targeting specific communities or organizations. One illustrative example is the "In Event of Moon Disaster" project by the MIT Center for Advanced Virtuality, which used deepfake techniques to depict former U.S. President Richard Nixon delivering a fictional speech about an Apollo 11 failure, demonstrating how synthetic media can convincingly simulate historical events (*Tackling the Misinformation Epidemic with "In Event of Moon Disaster,"* 2020). This example underscores the potential for deepfakes to distort historical narratives and fuel misinformation.

Automated Misinformation Through Multi-Modal Deepfakes: Advanced deepfake algorithms now integrate synthetic text, audio, and video to create highly persuasive disinformation campaigns tailored to specific audiences. Research indicates that multi-modal deepfakes exploit confirmation bias, increasing the likelihood of individuals believing and sharing manipulated content (Gambín et al., 2024). Public awareness remains low, as Europol reported that over 70% of people in the U.K. are unaware of deepfakes and their risks (European Union Agency for Law Enforcement Cooperation., 2024). Similarly, an iProov study found that only 0.1% of consumers in the U.S. and U.K. could consistently distinguish real from fake content. Older demographics are particularly vulnerable (*iProov Study Reveals Deepfake Blindspot*, 2025).

Financial and Corporate Scams: Cybercriminals increasingly use deepfakes to conduct sophisticated financial and corporate scams, often targeting businesses and high-net-worth individuals.

CEO/CFO Fraud: Deepfake technology is used to impersonate company executives, instructing employees to transfer funds or disclose confidential information. (Malinka, Firc, Šalko, et al., 2024) highlighted how deepfake-driven scams are becoming more sophisticated and harder to detect. Similarly, a case reported by Hong Kong authorities and covered in international media describes a finance employee at a multinational company who was deceived into transferring $25 million after fraudsters used deepfake technology to impersonate the firm's chief financial officer during a video conference call (Magramo, 2024).

Investment Scams: Deepfakes are employed to create fake endorsements by celebrities or trusted public figures, luring victims into fraudulent investment schemes (Scamwatch, 2024). These scams often exploit trust in trustworthy figures, making them particularly effective.

Non-Consensual Content, Blackmail and Reputation Sabotage: Our analysis of the literature stressed that deepfakes are increasingly weaponized to produce non-consensual content and facilitate blackmail schemes. This form of abuse disproportionately targets women and marginalized communities.

Non-Consensual Deepfake Pornography/ Sextortion: Deepfakes are used to create explicit content without the consent of the individuals depicted. (Blancaflor et al., 2024; Maras & Logie, 2024) identified technology-

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facilitated gender-based violence as a significant cybercrime enabled by deepfakes. These studies show that deepfake-generated intimate content is often used for revenge pornography and sextortion, causing significant emotional harm to victims. Attackers threaten victims with the release of fake explicit content unless a ransom is paid. (Flynn et al., 2022) highlighted how deepfake-driven image-based abuse disproportionately impacts marginalized groups, including LGBTQ+ individuals, young people, and racial minorities. Victims often face social stigma, discouraging them from seeking legal recourse.

Reputation Sabotage: Attackers use deepfake videos to damage reputations, posing significant risks in a digital world where online credibility is crucial. Over 90% of employers consider online search results in hiring decisions, with 77% of applicants facing negative outcomes due to damaging content (Horvitz, 2020; Mustak et al., 2023). Deepfakes amplify this threat by generating false yet convincing material, causing emotional distress and professional harm. For organizations, such attacks erode stakeholder trust, reduce business opportunities, and damage brands.

Bypassing Biometric Security Systems: Deepfake technology poses significant threats to biometric authentication systems, including facial recognition and voice verification used by banks, government agencies, and corporations. (S. Karim et al., 2024) introduced the MCGAN framework, demonstrating how adversaries could exploit deepfake technology to compromise biometric systems. Attackers can generate high-fidelity facial images and voiceprints to bypass identity verification, undermining the security of digital platforms. (R. Kumar et al., 2024) reported that deepfake videos used for identity verification often replicate facial movements and speech patterns with such accuracy that even advanced detection systems struggle to differentiate between genuine and synthetic inputs.

3.2 Thematic Analysis of Implications and Challenges

3.2.1 Theme 1: Emerging Trends in Cybercrime

Multi-Modal Deepfakes: Deepfake technology is advancing beyond single-modality attacks (video or audio) to multi-modal attacks that combine synthetic text, audio, and video. This integration creates highly convincing disinformation campaigns that are difficult to detect. (Gambín et al., 2024) emphasize how these multi-modal deepfakes can target victims across different platforms, making detection even more challenging. By combining video reenactment, voice cloning, and synthetic text generation, attackers can craft seamless, persuasive narratives for phishing scams, social engineering attacks, and disinformation campaigns.

Integration with Other Cyber Threats: Deepfake technology is increasingly integrated with other cyberattack vectors, amplifying the effectiveness of phishing schemes, social engineering tactics, and ransomware attacks. These hybrid attacks leverage synthetic media's realism to bypass traditional security measures and exploit human vulnerabilities, making them more dangerous, adaptable, and harder to detect (Flynn et al., 2022). The study highlights how attackers now use deepfake audio to impersonate executives or trusted individuals in phishing and social engineering schemes. The realistic nature of deepfake voice technology reduces suspicion, especially in high-pressure scenarios emphasizing urgency. Beyond phishing, deepfake technology extends to sophisticated ransomware attacks (Mustak et al., 2023). In these cases, attackers deploy deepfake videos to impersonate IT personnel or senior executives, instructing employees to download malicious files disguised as critical software updates or security patches. Once downloaded, the ransomware encrypts essential data and demands payment for its release. These attacks are particularly successful in industries with low cybersecurity literacy, where employees are more likely to trust directives from perceived authority figures.

Real-Time Deepfake Manipulation: Several studies highlighted the rapid advancement of real-time deepfake manipulation, enabling attackers to impersonate individuals during live video calls and virtual meetings (Frankovits & Mirsky, 2023; Kilger et al., 2021; Sharma et al., 2024). Unlike pre-recorded deepfakes, real-time versions use AI-driven facial reenactment and voice synthesis, making detection more difficult. This technology has serious cybersecurity implications. Real-time deepfakes pose national security risks as well. Threat actors could impersonate political leaders or officials to spread misinformation, disrupt operations, or manipulate diplomatic relations (Buckley et al., 2024). As deepfake tools become more accessible, even less sophisticated cybercriminals can exploit this technology.

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Audio Deepfakes: Audio deepfakes have become a significant cybercrime tool, using artificial intelligence to replicate voices with remarkable accuracy. Cybercriminals exploit this technology to impersonate individuals, bypass voice authentication, and conduct social engineering attacks (Barnett, 2023; Bharadwaj et al., 2023; Buckley, 2021; Cakir & Kasap, 2021; Novelli et al., 2024). Their effectiveness lies in the trust placed in voice communication, historically seen as more secure. Attackers generate convincing voice messages or calls to deceive victims into sharing sensitive information or transferring funds. As these forgeries become more sophisticated, even advanced security systems struggle to detect them, posing an increasing threat to individuals and organizations.

Synthetic Identity Fraud: Deepfake-generated synthetic identities are being used for fraudulent transactions, bypassing biometric security systems and identity verification protocols. (A. Kumar et al., 2024) report that attackers can create realistic personas with deepfake-generated facial features and voiceprints, allowing them to pass identity checks and gain access to financial accounts, government services, and corporate networks. This trend challenges organizations relying on biometric authentication as part of their security infrastructure.

AI-Enhanced Romance Fraud: Deepfakes are increasingly used in romance fraud schemes, where synthetic personas engage victims in long-term deception, often leading to financial exploitation. Several studies how attackers use deepfake-generated images, videos, and voice messages to build trust with victims, eventually manipulating them into transferring money or sharing sensitive information show (Choi et al., 2024; Cross, 2022; Lazarus et al., 2023; Thumboo & Mukherjee, 2024)). These scams often target vulnerable individuals, such as the elderly or socially isolated, making them particularly vulnerable.

Automated Disinformation Campaigns: Deepfake-driven misinformation campaigns are becoming more automated, with AI systems generating and distributing false narratives across social media platforms. A study notes that automation allows attackers to scale their operations, reaching larger audiences with minimal effort (Pawelec, 2022). These campaigns often target political processes, public health initiatives, and corporate reputations, undermining trust in digital communications and traditional media.

Manipulation of Legal Evidence: Deepfake technology is increasingly used to alter video and audio evidence, complicating legal investigations and judicial processes. A study highlights cases where deepfake-generated content has been introduced as evidence in court, raising questions about the integrity of digital evidence (Sandoval et al., 2024). This trend presents significant challenges for law enforcement and the legal system, requiring advanced forensic tools to detect synthetic media and verify digital evidence authenticity.

Deepfake-Driven Reputation Sabotage: Attackers are leveraging deepfakes to damage corporate or individual reputations by creating false narratives or compromising content. Deepfake-driven reputation sabotage can lead to emotional distress, professional harm, and financial loss. One study even shows how deepfakes contribute to misinformation, fraud, and reputation damage, emphasizing the societal and technological risks posed by this technology (Boutadjine et al., 2023). It also illustrates cases where manipulated media has been used to defame individuals, spread false accusations, and undermine the credibility of organizations. These attacks often exploit the victim's public image, making it difficult to refute false claims once the content spreads across social platforms. The rising accessibility of deepfake creation tools has amplified the risk of reputation sabotage, as malicious actors can easily generate compromising videos or audio recordings without requiring significant technical expertise.

3.2.2 Theme 2: Ethical, Legal, and Societal Implications

Ethical Implications: The ethical implications of deepfakes extend beyond their use in cybercrime, touching on issues of privacy, consent, and the broader societal impact of synthetic media. (Brown, 2023) discusses how the creation of synthetic media without consent violates individual privacy and autonomy, raising questions about the ethical boundaries of AI research and the responsibilities of developers. Deepfakes can be weaponized to create non-consensual explicit content, manipulate political discourse, or fabricate evidence, undermining trust in digital communication. One particularly concerning trend involves the use of deepfakes in non-consensual pornography and gender-based abuse, disproportionately targeting women. In the UK, for example, individuals have been sent to prison for creating and distributing deepfake pornography involving real people, highlighting the growing legal response to this harmful practice (*Government Cracks down on 'Deepfakes' Creation*, n.d.). Maras and Logie highlight the psychological harm and reputational damage faced by victims,

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emphasizing that the ease of creating such content with publicly available deepfake tools exacerbates the ethical dilemma (Maras & Logie, 2024). Similarly, studies highlight the dual-use dilemma of deepfake technology, where its beneficial applications in education, entertainment, and accessibility are overshadowed by its potential for harm (Mustak et al., 2023). The ethical challenge extends to developers of deepfake technology, who face the responsibility of ensuring that their innovations are not misused. Studies also stress the need for ethical governance frameworks, including ethical training for AI developers, to promote responsible innovation and minimize harm (Pawelec, 2024).

Legal Frameworks: Legal frameworks addressing the misuse of deepfakes remain fragmented and inconsistent across jurisdictions. Some studies have argued that existing laws fail to account for the cross-border nature of cybercrimes involving deepfakes, complicating efforts to hold perpetrators accountable (Brown, 2023; Chawki, 2024; Flynn et al., 2022). This challenge is further complicated by the rapid pace of technological advancement, which often outstrips legislative processes. Studies have called for the development of international treaties that harmonize legal responses to deepfake-related crimes(E. Blancaflor et al., 2023; G. Buckley et al., 2024). Such frameworks must balance the need for regulation with the protection of free expression and innovation. The lack of cohesive legislation creates barriers for victims seeking legal recourse, especially in cases of non-consensual deepfake content or reputation sabotage. Deepfakes also threaten the integrity of digital evidence, as falsified content has been used to mislead legal proceedings (Sandoval et al., 2024). Verifying authenticity now requires advanced forensic tools, adding pressure to already overburdened legal systems.

Societal Implications: Data collected in our SLR highlighted that the societal implications of deepfakes are profound, particularly regarding trust in digital communication, democratic processes, and social cohesion. (Flynn et al., 2022) note that the mere possibility of media manipulation erodes public confidence in information sources, even when no deepfakes are involved. This phenomenon, known as the "liar's dividend," enables individuals to dismiss authentic evidence as fake, further undermining trust in legitimate information (Brown, 2023). Deepfakes also threaten democratic processes by enabling the creation of false narratives and misleading political content. These studies explore how deepfakes can undermine trust in electoral systems and political discourse. For instance, deepfake videos depicting political candidates making inflammatory statements can influence voter behavior, potentially swaying election outcomes. Deepfakes also contribute to social polarization by fueling disinformation campaigns and amplifying societal divisions. (Pawelec, 2024) highlights how automated disinformation campaigns, powered by synthetic media, can target specific communities with tailored narratives, exacerbating existing tensions and spreading falsehoods. Additionally, vulnerable populations, including the elderly and those with limited digital literacy, are more susceptible to deepfake-driven scams and misinformation (Malinka, Firc, Šalko, et al., 2024). This digital inequality further widens the gap between those who can critically assess online content and those who cannot, increasing the risk of exploitation.

3.2.3 Theme 3: Identified Research Gaps

Despite the growing research on deepfakes and their impact on cybersecurity, our research found that several critical gaps remain. These gaps slow progress in developing effective defense strategies, leaving individuals, organizations, and societies more vulnerable to evolving threats.

Detection Limitations and Evasion Techniques: While significant advancements have been made in deepfake detection, existing frameworks remain reactive rather than proactive. Juefei-Xu et al. (2022) emphasize the ongoing arms race between deepfake generation and detection, with GANs continuously evolving to bypass current detection systems (Juefei-Xu et al., 2022). Another study highlights the need for improved detection algorithms capable of identifying low-resolution deepfakes, which are often used in misinformation campaigns and phishing attacks (Malinka, Firc, Šalko, et al., 2024). A key research gap involves the development of multi-modal detection systems that can analyze cross-platform synthetic content. A study argues that current detection tools are largely optimized for single-modality content, such as video or audio, making it difficult to detect composite attacks that combine video, audio, and synthetic text (Asha et al., 2024). Moreover, studies stress the need for scalable detection solutions that can operate in real-time across different media formats (N. A. Karim et al., 2024). Research shows that proactive detection measures, such as digital watermarking and

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cryptographic signatures embedded during content creation, show promise but remain underexplored ("Enhancing Deepfake Detection," 2025; Flynn et al., 2022; Nadimpalli & Rattani, 2024). They emphasize the need for further research into authentication protocols that can verify the integrity of digital media at the point of creation. Additionally, AI-driven behavioral analysis offers potential for detecting deepfake-related anomalies in communication patterns, but current models are limited in their accuracy and scalability.

Legal and Regulatory Frameworks: The fragmented nature of deepfake-related legislation presents a significant challenge for governance and enforcement. Many studies identify gaps in international frameworks addressing non-consensual deepfakes and technology-facilitated gender-based violence (Batool et al., 2024; De Angeli et al., 2023; Falduti & Griffo, 2022; Kira, 2024; Viola & Voto, 2023). Other studies further highlight the absence of comprehensive regulations in many countries, emphasizing the need for policies that balance free expression with protection against digital manipulation (Maras & Logie, 2024; Songja et al., 2024). A recent study points to the lack of harmonization between the U.S., EU, and China regarding deepfake regulations, complicating enforcement and cross-border cooperation (Zheng et al., 2025). This legal fragmentation leaves victims with limited recourse, particularly when deepfake content is disseminated across jurisdictions. Many country and region specific studies also advocate for international treaties to standardize legal responses and improve victim protection (Blancaflor et al., 2024; Dey & Dey, 2025; Franco et al., 2022; Piyatumrong, 2024). Another critical gap lies in the attribution of liability for deepfake-generated harm. Current frameworks often fail to clarify whether accountability lies with the content creator, platform, or distributor (de Ruiter, 2021; Köbis et al., 2021; Pawelec, 2024). This ambiguity complicates efforts to hold perpetrators accountable and provide restitution to victims. Further research is needed to explore how liability can be appropriately assigned while safeguarding innovation and freedom of expression. Additionally, existing frameworks often lag behind technological advancements, leaving victims without adequate protection. Moreover, legal protections for businesses against deepfake-driven deception are limited. In a review of legal frameworks across several countries, many studies found that most jurisdictions lack specific regulations addressing deepfake-driven cybercrime, leaving businesses and individuals vulnerable to reputational sabotage and financial losses (Arshed et al., 2024; de Rancourt-Raymond & Smaili, 2023; Kurshan et al., 2024).

Human-Centered Detection and Awareness: Research has shown that while technological advancements are crucial, human-centered approaches to deepfake detection and awareness remain underexplored (Malinka, Firc, Šalko, et al., 2024). They stress the importance of understanding how users perceive and detect deepfake content, particularly in voice phishing (vishing) scenarios. Other literature highlights gaps in fraud prevention messaging, noting that existing strategies are outdated and ineffective against synthetic media attacks (Cross, 2022; Matli, 2024). There is also limited research on the psychological impact of deepfake-enabled social engineering attacks (Bendiab et al., 2025; Schmitt & Flechais, 2024). Employees targeted by deepfake-driven scams may experience guilt, shame, and diminished trust in their workplace. Studies emphasize the need for interdisciplinary research examining how organizations can provide psychological support and resilience training for affected employees (Alanazi et al., 2025; Caci et al., 2024; Vecchietti et al., 2025). Moreover, public awareness campaigns often focus on general cybersecurity practices without addressing the specific risks posed by deepfakes. Research is needed to develop targeted educational programs that equip users with the skills to identify synthetic content and respond appropriately.

Ethical and Societal Implications: Ethical considerations surrounding deepfakes extend beyond privacy violations to include issues of consent, identity manipulation, and misinformation. Studies have highlighted the limited consideration of developer ethics in synthetic media creation, calling for frameworks that embed ethical practices into the development lifecycle (Bowen, 2024; Pawelec, 2024). They also stress the need for further research into how deepfakes influence societal trust, particularly in regions with limited media literacy as well as in high-stress or unpredictable situations where individuals are more likely to rely on confirmation bias, motivated reasoning, and other cognitive shortcuts (Songja et al., 2024). These studies further identify gaps in understanding the psychological and emotional toll of deepfake attacks, particularly in cases of blackmail, harassment, and non-consensual pornography (Blancaflor et al., 2024; De Angeli et al., 2023; Falduti & Tessaris, 2023; Šepec & Lango, 2020). Victims often experience long-term trauma and reputational harm, yet empirical data on victim coping mechanisms and support systems remains scarce. The ethical implications of synthetic

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identity fraud are also underexplored. As deepfake technology enables the creation of realistic yet fictitious personas, questions arise regarding the boundaries of digital identity creation and the potential for exploitation.

Attribution and Accountability: Accurately attributing deepfake attacks remains a significant challenge due to the anonymous nature of synthetic content creation and dissemination. Researchers have emphasize the need for improved attribution frameworks that can trace the origins of deepfake-generated media, particularly in open-world scenarios (Sun et al., 2025). Current forensic tools often lack the sophistication required to distinguish authentic content from highly realistic deepfakes. They highlight the need for standardized forensic methodologies to verify the authenticity of digital evidence in legal contexts (Sandoval et al., 2024). Without reliable attribution mechanisms, victims of deepfake-driven disinformation often struggle to prove their innocence, complicating efforts to seek justice. Further research is needed to develop forensic tools capable of analyzing synthetic content across platforms and identifying patterns indicative of deepfake manipulation. Additionally, legal frameworks must evolve to ensure that false accusations stemming from deepfake-generated content are addressed effectively.

Emerging Technologies: Emerging technologies, such as blockchain, offer promising solutions for content authentication, yet their implementation remains experimental. Researchers have advocated for further exploration of blockchain-based frameworks that can provide immutable records for verifying media authenticity (Gambín et al., 2024). Such systems could enable users to trace content origins and identify manipulated media more effectively. They also identify gaps in multi-modal detection models capable of analyzing cross-platform attacks (Asha et al., 2024). Current frameworks are predominantly optimized for single-modality detection, leaving multi-modal deepfake content. They combine synthetic video, audio, and text which are largely undetected. Additionally, AI-driven detection systems remain underexplored. While AI models can enhance detection accuracy, they also raise ethical concerns regarding surveillance and privacy. Further research is needed to develop AI-based detection frameworks that balance security with individual rights.

Cross-Cultural and Regional Insights: This review has uncovered the lack of region-specific studies examining the cultural and social impact of deepfakes, emphasizing the need for localized approaches to regulation and education (Songja et al., 2024). Most existing research is concentrated in North America, Europe, and parts of Asia, overlooking the unique challenges faced by underrepresented regions such as Africa, South America, and the some parts of the Middle East. In many developing regions, rapid digital adoption has outpaced the development of cybersecurity frameworks, creating vulnerabilities that cybercriminals can exploit. These studies argue that regional differences in technological infrastructure, cybersecurity policies, and socio-political dynamics shape the impact and effectiveness of deepfake attacks (Blancaflor et al., 2024). Further research is needed to explore how deepfakes are weaponized in these regions and how policymakers can develop region-specific defense strategies. Moreover, the lack of legal frameworks addressing deepfake threats in underrepresented regions leaves individuals, businesses, and political figures particularly vulnerable. Victims often face significant challenges in seeking justice due to limited resources, weak enforcement mechanisms, and a lack of technical expertise among law enforcement agencies.

Demographic Susceptibility to Deepfake Attacks: Understanding demographic susceptibility to deepfake-driven cyberattacks remains a critical research gap. Older populations often struggle to recognize synthetic media, increasing their risk of impersonation scams (Ebner et al., 2023; Mustak et al., 2023). Limited access to cybersecurity education also heightens vulnerability among lower socio-economic groups. Deepfake attacks disproportionately impact women, LGBTQ+ individuals, and racial minorities, yet research has largely focused on non-consensual deepfake pornography and harassment (Kira, 2024). Additionally, employees in finance, HR, and IT face heightened risks of deepfake-driven social engineering, but little research examines how job roles influence targeting. The psychological impact on victims, including emotional distress and reputational harm, remains underexplored. Addressing these gaps is crucial for developing targeted awareness programs and cybersecurity interventions.

Taken together, these findings provide the empirical and conceptual foundation for the interpretive analysis presented in the following section, where we synthesize cross-cutting patterns, examine their implications, and outline priorities for future research and practice.

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4 DISCUSSION, FUTURE RESEARCH DIRECTIONS, AND RECOMMENDATIONS

The findings of this systematic literature review provide insights into how cybercriminals exploit deepfakes to conduct cyberattacks, examined through direct responses to the three RQs guiding this review. A key pattern emerging from this analysis is that deepfake-enabled cyberattacks remain structurally under-researched relative to the scale and speed of the threat. Across the 188 studies reviewed, most academic attention is concentrated on detection algorithms, while far less work addresses attacker behaviors, victim experiences, organizational vulnerabilities, or the broader socio-technical systems in which deepfakes operate. This imbalance reveals a misalignment between where research efforts are focused and where real-world risks continue to escalate. As generative AI tools make deepfake creation increasingly accessible and trivial, the threat landscape is evolving more rapidly than the research ecosystem designed to understand and mitigate it.

RQ 1, which investigates emerging trends in cybercrime involving deepfakes, is addressed after an analysis of how cybercriminals leverage this technology to conduct increasingly sophisticated attacks. Deepfake technology has rapidly evolved into a scalable and adaptable cybercrime tool, enhancing social engineering, financial fraud, and disinformation campaigns. Advances in artificial intelligence, particularly GANs, have made deepfakes faster, cheaper, and more accessible, increasing their use in phishing, social engineering, ransomware, and BEC scams. (Juefei-Xu et al., 2022; Malinka, Firc, Šalko, et al., 2024). Despite advancements in deepfake detection, existing systems remain largely reactionary and struggle to keep pace with evolving synthetic media techniques. Most rely on inconsistencies in facial movements, audio artifacts, and image anomalies, but some studies noted that low-resolution deepfakes can evade detection (A. Kaur et al., 2024; Köbis et al., 2021; Malinka, Firc, Šalko, et al., 2024). Attackers are able to impersonate individuals in real-time video calls, bypassing authentication processes and leading to fraudulent transactions. Multi-modal deepfakes, which integrate synthetic video, audio, and text, present additional challenges by making manipulated content more convincing and harder to detect. The rise of deepfake-enhanced phishing attacks have become a major concern, as cybercriminals use AI-generated voices and faces to deceive employees into authorizing payments or disclosing sensitive data (Damiani, n.d.; Magramo, 2024). Many studies emphasize that deepfake-driven cyberattacks undermine verification processes, especially in remote work environments where identity authentication relies heavily on digital communication. These developments require more adaptive cybersecurity strategies to mitigate risks (Gambín et al., 2024).

Collectively, these patterns indicate that offensive innovation in synthetic media is advancing more quickly than defensive capacity, creating a growing asymmetry between attacker capabilities and the readiness of organizations and individuals to detect and resist deepfake-enabled deception.

RQ 2 is addressed by exploring how deepfakes erode trust in digital communication, contribute to misinformation, and complicate legal proceedings. The "*liar's dividend*" phenomenon allows malicious actors to dismiss authentic evidence as fabricated, undermining credibility in media and digital communication (Pawelec, 2024). This has major implications for legal proceedings, where deepfake-generated evidence can obstruct forensic investigations (Afroz et al., 2025; Łabuz & Nehring, 2024; Sandoval et al., 2024), and for politics, where synthetic media has fueled disinformation campaigns, such as the 2019 Gabon coup attempt (Cahlan, 2020). Deepfake-enabled reputation sabotage is also on the rise. Attackers can fabricate compromising content to manipulate public perception, harming individuals and organizations. Gender-based abuse and non-consensual deepfake pornography continue to rise as well, yet legal protections remain inadequate. Additionally, over 90 percent of employers rely on online search results in hiring decisions, amplifying the risks of deepfake-driven blackmail and misinformation (Horvitz, 2020). The dual-use nature of deepfake technology further complicates regulation, as its legitimate applications in education and entertainment must be balanced with cybersecurity protections. Addressing these threats requires stronger detection frameworks, international collaboration, and crisis management strategies (Maras & Logie, 2024; Songja et al., 2024).

These findings show that deepfakes function not only as technical artifacts but as powerful socio-psychological tools that destabilize evidentiary standards, exploit cognitive vulnerabilities, and weaken the institutional and social foundations of trust.

RQ 3, which explores research gaps and challenges in understanding and addressing deepfake-enabled cyberattacks, is addressed by identifying key shortcomings in current research and detection efforts. Despite growing academic interest in deepfake cyberattacks, major gaps remain in empirical research and cross-

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disciplinary studies. There are significant disparities in the distribution of deepfake-related risks, with certain demographic groups and geographic regions disproportionately affected. They noted that older adults, women, LGBTQ+ individuals, and racial minorities are more vulnerable to deepfake-driven cyberattacks, including non-consensual pornography and online harassment (Ebner et al., 2023; Kira, 2024; Review & Mengesha, 2024). There is also a lack of research on deepfake-related cybercrime in underrepresented regions where cybersecurity infrastructure is less developed.

Limited real-world case studies exist, with many focusing on theoretical frameworks rather than documented cyberattacks. Job-specific vulnerabilities are also underexplored, with employees in finance, human resources, and IT facing heightened risks of deepfake-driven fraud and social engineering. Furthermore, psychological impacts on victims remain inadequately studied, with limited data on emotional distress, reputational harm, and social consequences. These shortcomings restrict the development of targeted cybersecurity interventions, regulatory responses, and forensic tools.

Taken together, this evidence highlights a systematic imbalance in the literature: technical detection research receives disproportionate attention, while socio-technical dynamics, victim experiences, regional disparities, and organizational practices remain comparatively underexplored.

While existing studies are making important progress, several areas require deeper investigation to strengthen the field's long-term resilience against deepfake-enabled cybercrime. First, future research should examine how deepfakes interact with broader GenAI ecosystems, including automated social engineering, chatbot-driven impersonation, and hybrid attacks that combine synthetic media with large language model outputs. Understanding these convergences will help identify compound threat patterns that are not yet well-documented in the literature.

Second, more attention is needed on integrated defensive architectures rather than isolated detection tools. Future studies should explore how provenance technologies, behavioral analytics, multi-factor verification, and anomaly monitoring can function together as part of coordinated organizational defense frameworks. This includes examining system-level interactions and evaluating the resilience of these tools when adversaries deploy multimodal synthetic media.

Third, organizational readiness remains underexplored. Research should analyze how different industries update verification procedures, incident response plans, and internal controls in response to synthetic impersonation risks. This includes assessing how organizations train employees, redesign workflows, and adopt governance strategies specifically tailored to deepfake threats.

Finally, interdisciplinary research is critical for understanding how deepfake attacks influence social trust, political stability, and psychological well-being. Future studies drawing from psychology, communication, law, human–computer interaction, and international relations will help contextualize deepfake risks within broader social and geopolitical systems.

The review also highlights key recommendations to address these challenges. Effectively countering deepfake-driven cyberattacks requires technological innovation, regulatory alignment, and public awareness initiatives. AI-based detection systems must adapt to multi-modal deepfakes by integrating cross-modal analysis of video, audio, and text (Aarthe et al., 2024; Alkaabi et al., 2023; Bendiab et al., 2025; S. Karim et al., 2024). Researchers recommend embedding digital watermarking and cryptographic signatures into media content to improve authentication verification. Blockchain-based media authentication could further enhance content integrity and provenance tracking (Flynn et al., 2022; Gambín et al., 2024). From a regulatory perspective, a standardized, international legal framework for deepfake detection, reporting, and prosecution is critical (Zheng et al., 2025). Stronger cross-border cooperation is needed to improve enforcement mechanisms and hold perpetrators accountable. Industry-academic collaborations could further establish best practices for detection and mitigation. Raising public awareness is equally crucial. Digital literacy programs tailored to high-risk groups, including older adults and marginalized communities, could improve resilience against deepfake deception. Research has already shown that individuals exposed to awareness campaigns are significantly better at detecting synthetic content (Malinka, Firc, Šalko, et al., 2024). Ethical AI development must be governed by standards that promote transparency, accountability, and harm mitigation. Industry-wide practices, such as labeling synthetic content and embedding provenance mechanisms, would help prevent misuse while preserving legitimate applications. Expanding interdisciplinary research is essential to

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understanding the psychological, societal, and geopolitical dimensions of deepfake-enabled cybercrime. Collaboration between cybersecurity experts, legal scholars, and policymakers could lead to more effective countermeasures (Review & Mengesha, 2024; Thumboo & Mukherjee, 2024).

Recent studies in adjacent areas reinforce the urgency of emerging AI-related cyber risks. Research on AI-enabled vulnerabilities in software bills of materials (SBOMs) demonstrates how AI introduces new exploit pathways across large software supply chains, while work on AI-security in IoT ecosystems highlights how model-driven attacks can amplify existing device-level weaknesses (Radanliev, De Roure, et al., 2024; Radanliev, Santos, et al., 2024). Although these studies do not focus on deepfakes specifically, they illustrate a broader pattern in which generative AI expands the cyber-attack surface. Our review complements these developments by showing how synthetic media introduces a parallel set of socio-technical risks, particularly in deception-based attacks, impersonation, and trust exploitation.

As deepfake technology advances, cybercriminal tactics will continue to evolve, necessitating proactive security measures, stronger regulations, and cross-sector collaboration. A comprehensive response integrating technological innovation, legal accountability, public education, interdisciplinary research, and public-private partnerships is essential for mitigating deepfake-enabled cybercrime and safeguarding digital trust.

5 CONCLUSION

This review situates deepfake-enabled cyberattacks within broader debates in cybersecurity about the increasing sophistication, scalability, and personalization of digital threats. Deepfake technology exemplifies how advances in artificial intelligence can simultaneously create new opportunities and generate novel vectors for harm. By synthesizing evidence from 188 studies, the review shows how synthetic media is being weaponized across multiple attack types, identifies where current defenses fall short, and highlights the socio-technical nature of the risks involved. These insights provide a foundation for strengthening both technical controls and governance responses as the threat landscape continues to evolve.

The findings demonstrate that deepfakes are rapidly becoming a preferred tool for cybercriminals, enhancing the effectiveness of phishing, social engineering, identity fraud, disinformation campaigns, and reputation sabotage. As deepfake technology and GenAI systems continue to evolve, the threat landscape is likely to become more complex, which will require innovative detection mechanisms, improved organizational controls, and more coherent legal and regulatory frameworks. Addressing these challenges will require coordinated action from researchers, industry practitioners, policymakers, and civil society. The review provides an evidence base that can support this work, informing both future academic research and the design of practical strategies for countering deepfake-enabled cybercrime.

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APPENDIX

List of 188 Papers included in the Systematic Literature Review

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